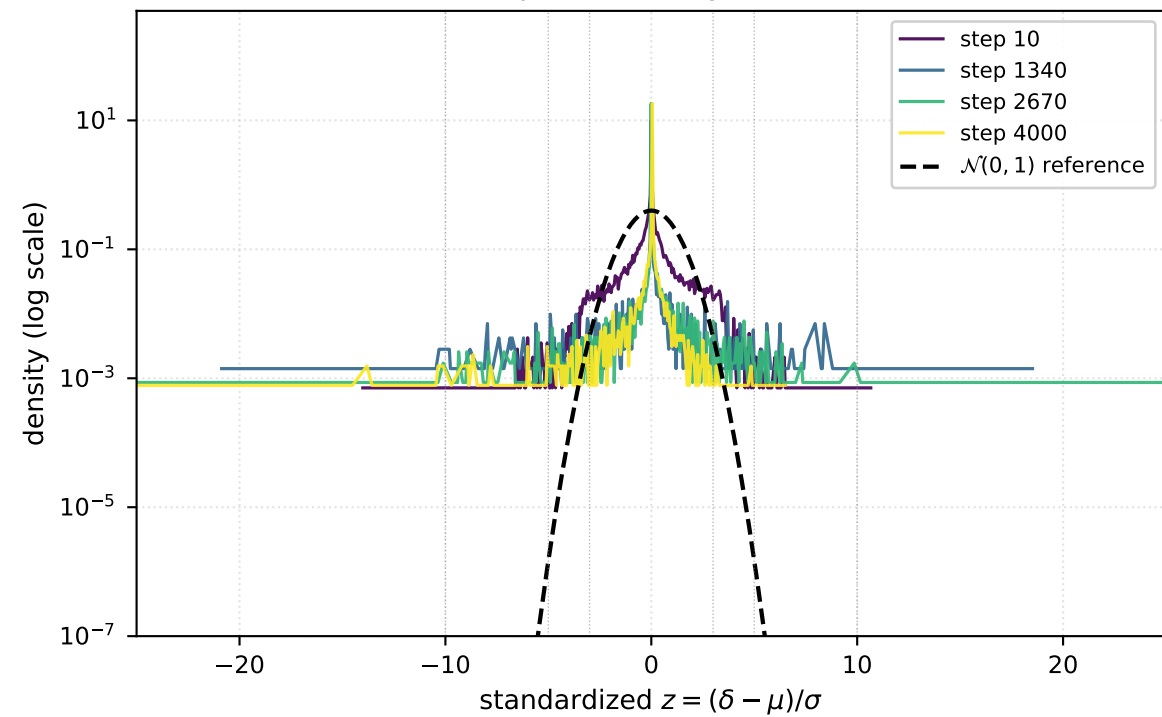
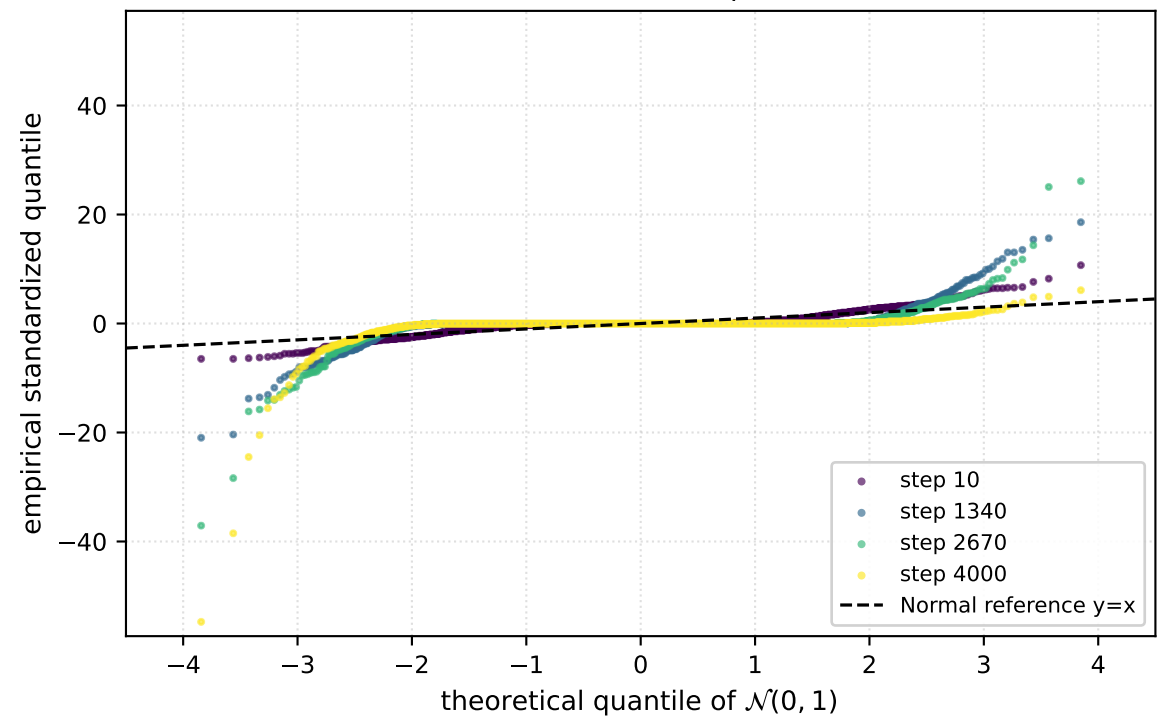


Heavy-tail behaviour of token-level $\delta = \log \pi_{\theta} - \log \pi_{\theta_{\text{old}}}$ in GSPO training

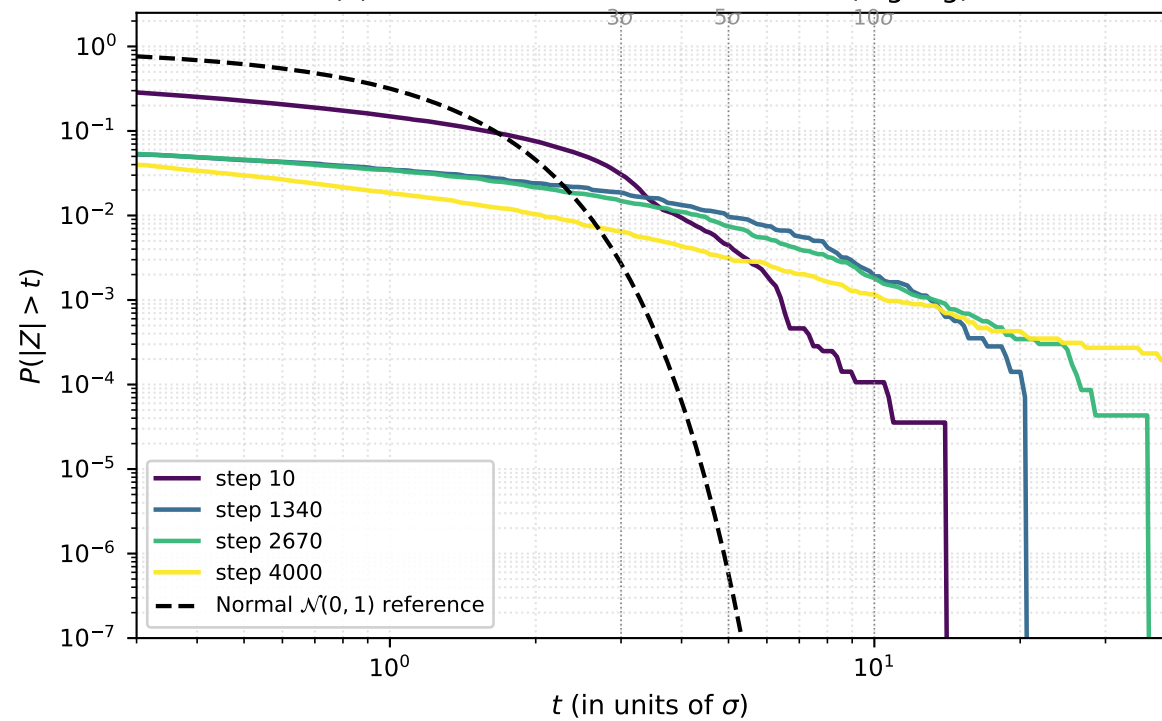
(a) Empirical density vs $\mathcal{N}(0, 1)$



(b) Normal Q-Q plot



(c) Tail survival function vs Normal (log-log)



(d) Heavy-tail metrics over training

